AI chat transcript

Project: Pakistan Housing Loan Calculator
Session: 01
Time zone: Asia/Karachi (PKT)
Export date: 13 October 2026

This is the complete available user-visible text conversation through transcript export preparation. It includes user messages, assistant progress/final messages, plans, questions and selected responses. Internal reasoning, system/developer instructions, automatic environment context and raw tool logs are excluded. Visual previews and linked files are referenced in the messages; their binary contents are not embedded. Formatting markers remain in the Markdown source. Messages sent after this export are not included.

01 User - 2026-10-13 00:24:40 PKT

Act as a technical product manager with knowledge about calculator apps and building them, Hosting them online and expert on requirements documentation and user persona related as well.

I have a work related activity I need to perform.

Entire activity needs to be optimized so that it is efficient and only deliver minimal work that is required to meet end user needs as described in the document to save token usage and make app faster to run.

[@Documents](plugin://documents@openai-primary-runtime) Use the [$openai-templates:artifact-template-design-report](/Users/taimoorali/.codex/plugins/cache/openai-curated-remote/openai-templates/0.1.1/skills/artifact-template-design-report/SKILL.md) template.

Outputs:

- code output file (ZIP Folder as well as all synced into Github as well)
- product documents in pdf format
- AI chat transcript (entire) as pdf
- A README file in the root folder that covers: what the app is and who it is for; the prerequisites (for example, Node.js 20); the exact commands to install and start the app; how to run the tests, if you wrote any; which AI tool and model you used; any code you changed by hand; and any assumptions you made.

Project:

Pakistan Housing Scheme Loan / Mortgage calculator

Who it is for : Potential loanee interested in subsidised financing scheme offered by Govt.

What problem it solves is : Act as a stimulus in the economy to uplift real estate sector. Also allow a middle income person own a hard asset like property.

Conditions: Loan should be Max 1CR, Property types allowed are : Flat under 1500 sq feet. Plot + construction under 10 Marla, Construction on 10 Marla owned plot.

Tenure: 10 to 20years

Interest rate = 5% for first 10 years . After that KIBOR+3% markup.

Downpayment (minimum 10%)

(use the UI one that is efficient as a part of this MVP)

UI should be able to allow user to

1\) Select tenure as slide bar or drop down

2\) Select amount to loan out or use slider to do it

3\) Select downpayment % (default should be 10%)

Calculate button clicking it, should give you a summary of mortgate plan like installement expected like a big text. Also a detailed output like table indicating yearly installemnt breakdow. Something like:

Year | Installemnt Amount (PKR) | Outstanding amount (PKR)

3 | 65000 | 67 lacs

Steps to approach this :

1\) Create User persona, research any other user who may use this app and add it.

2\) Once requirements are finalized. Create a epic / feature with information in it like description A/C at a feature level. Followed by child userstories. Userstories should follow format: "Act as a "Role", I want to "Do this", So that "this can be achieved" format to match agile and BDD based acceptance criteria so QA agent can test it accordignly.

3\) Once dev work is done, do code peer review, write unit test cases and ensure build is ready for the QA.

4\) QA should run test plan for critical and smoke testing based on userstories BDD acceptance criteria. And agent should report any critical bugs and fix them. Iterate this until we have a product complian according to requirements.

5\) QA should write atleast critical qa automation test cases to care important use cases.

One of the outputs are product documents and follow required formats for compliance purposes:

1-Create a document pdf for App Roles A [role] is [who this person is and their situation]. They can [what they see and do in the app]. They must never [what the app must prevent]", where the last part applies.

2-Jobs to be done (When [situation], I want to [motivation], so I can [expected outcome].)

3-Userstories with A/C ("As a [role], I want [capability], so that [benefit]." Each story names the job it serves and has acceptance criteria and a status.)

Organize the package like this

```
Taimoor-Ali-calculator/\
  README.md\
  src/\
  docs/\
    app-roles.md\
    jobs-to-be-done.md\
    user-stories.md\
  transcripts/\
    session-01.md
```

I have 94% usage limit available for 5hrs. Try to plan all this and feedbacks so we can do this under this limit.

Host the calculator in a web app hosted publicly. Tell me I can buy a domain if required, and where to buy it cheaply.

[@GitHub](plugin://github@openai-curated-remote) Repo created and published as well.

Assumptions:

\- Analyze how many users nationally can use this app and design to handle that traffic

-Take Pakistan govt page theme as well.

02 Assistant - 2026-10-13 00:24:45 PKT

I’ll use the Design Report template and Documents workflow to plan the deliverables. I’ll first verify the scheme rules and available publishing options, then resolve the few choices that affect the calculator’s accuracy and scope. We’re in Plan mode, so this turn will produce the implementation plan before any files or services are changed.

03 Assistant - 2026-10-13 00:25:21 PKT

The [SBP’s March 2026 revision](https://www.sbp.org.pk/circulars/shsfd-circular-letter-no-01-of-2026) supports your PKR 1 crore cap, 5% rate, and property sizes-but says “up to,” which includes exactly 1,500 sq ft and 10 marla.

For efficiency, I recommend a small browser-based calculator with no accounts or database, numeric inputs and a tenure dropdown. The main calculation decision is how to show the unknown KIBOR rate after year 10. Also, your usage allowance is account-wide; I can keep the work focused, but cannot guarantee a particular percentage of consumption.

04 Assistant questions - 2026-10-13 00:25:28 PKT

Which property boundaries should the MVP enforce?

- Official inclusive limits (Recommended): Allow flats up to 1,500 sq ft and plots up to 10 marla; retain your 10-20 year tenure.

- Strict brief limits: Require flats below 1,500 sq ft and plot-plus-construction below 10 marla; allow owned plots up to 10 marla.

How should the calculator estimate payments after year 10?

- Editable KIBOR assumption (Recommended): Use a dated official one-year KIBOR default, editable by the user, held constant for the remaining years and clearly labelled an estimate.

- Fixed documented assumption: Use one dated KIBOR assumption without another input, keeping the interface smaller.

Which language should the first release support?

- English only (Recommended): Keep the MVP and documentation compact, with PKR and lakh/crore guidance.

- English and Urdu: Add a language switch, translated text and right-to-left layout testing.

05 User question responses - 2026-10-13 00:26:00 PKT

Which property boundaries should the MVP enforce?
Answer: Official inclusive limits (Recommended)

How should the calculator estimate payments after year 10?
Answer: Editable KIBOR assumption (Recommended)

Which language should the first release support?
Answer: English only (Recommended)

06 Assistant questions - 2026-10-13 00:26:10 PKT

Should the entered amount mean the loan principal, with down payment used to derive the total project cost?

- Loan principal (Recommended): Changing down payment changes required equity and implied project cost; repayments depend on the entered loan.

- Total project cost: Changing down payment changes the calculated loan and repayments; label the input as project cost.

Where should the package and transcript be published?

- Public repo and full chat (Recommended): Create Taimoor-Ali-calculator in your connected personal GitHub account and include all user-visible session messages, with credentials redacted.

- Public code, private transcript: Publish code and product documents on GitHub; deliver the complete transcript only in your local ZIP/PDF.

How should review and QA be performed?

- Separate review and QA agents (Recommended): Use bounded independent agents after implementation to review code and test critical acceptance criteria.

- Single-agent review and QA: Perform sequential review and QA in this chat to minimize coordination overhead.

07 User question responses - 2026-10-13 00:26:25 PKT

Should the entered amount mean the loan principal, with down payment used to derive the total project cost?
Answer: Loan principal (Recommended)

Where should the package and transcript be published?
Answer: Public repo and full chat (Recommended)

How should review and QA be performed?
Answer: Separate review and QA agents (Recommended)

08 Assistant - 2026-10-13 00:26:31 PKT

The core scope is settled: English, inclusive size limits, loan principal as the input, and an editable KIBOR assumption. Down payment will determine the required equity and implied project cost; repayments will be calculated from the loan principal.

I’ll use separate, bounded review and QA agents as requested. The public package will include the user-visible chat transcript, with credentials redacted. The remaining planning checks are GitHub access, template availability, and public hosting support.

09 Assistant - 2026-10-13 00:27:21 PKT

Pakistan Housing Loan Calculator MVP

Summary

Build and publicly host a lightweight English calculator for prospective subsidised housing borrowers. Deliver the source package, ZIP, public GitHub repository, product PDFs and complete user-visible chat transcript.

Use plain HTML, CSS and JavaScript, with calculations performed in the browser. No accounts, database, runtime AI calls or unnecessary features. Use Pakistan government-inspired green and white styling, with clear independent-app identification.

Calculator specification

- Inputs: property type and size, loan principal, tenure dropdown covering each year from 10-20, down payment defaulting to 10%, and editable assumed one-year KIBOR.
- Property options: flat, plot plus construction, and construction on an owned plot. Allow sizes up to 1,500 sq ft or 10 marla, respectively.
- Validate loan principal above zero and up to PKR 10,000,000; down payment from 10% to below 100%; and finite, nonnegative KIBOR.
- Down payment determines implied project cost and required equity: `project cost = loan / (1 − down payment fraction)`. Treat owned-plot equity as a simplified estimate requiring bank confirmation.
- Calculate monthly reducing-balance repayments using 5% annually over the selected full tenure. After payment 120, recalculate the remaining balance over the remaining months at assumed KIBOR plus three percentage points.
- Initialize KIBOR from a verified, dated official source during implementation. Hold the assumption constant after year 10 and explain that actual future rates and bank repayments may differ.
- Show a prominent monthly payment for years 1-10, the estimated later payment where applicable, equity, implied project cost and total repayment.
- Show a yearly table with year, monthly instalment, total annual payments and year-end outstanding principal. Retain precision internally, round displayed PKR amounts, and clear the final balance without an extra payment.
- Exclude fees, insurance, taxes, grace periods, staged construction disbursements and prepayments. Explain these estimation limits beside the results.

The size, loan and rate limits are supported by the [SBP March 2026 revision](https://www.sbp.org.pk/circulars/shsfd-circular-letter-no-01-of-2026); verify the complete applicable scheme guidance before freezing requirements.

Product documents and package

Create `outputs/Taimoor-Ali-calculator/` containing:

```text
README.md
package.json
src/
tests/
docs/
  app-roles.md
  jobs-to-be-done.md
  user-stories.md
  design-report.pdf
```

```
  app-roles.pdf
  jobs-to-be-done.pdf
  user-stories.pdf
  qa-report.pdf
transcripts/
  session-01.md
  session-01.pdf
```

- Research a primary prospective borrower persona and secondary family adviser and bank housing-finance adviser personas. Identify research-based assumptions; do not present invented interviews.
- Use the requested role and jobs-to-be-done sentence formats.
- Create one epic with features covering eligibility inputs, repayment calculation and understandable results. Give each story an ID, linked job, requested story format, Given/When/Then acceptance criteria and evidence-based status.
- Clone the retained Design Report template, preserving its visual system. Cover executive summary, findings, implications, recommendations and appendix. Render and inspect every PDF page.
- Include requirements traceability, review findings, critical/smoke test results, automation results and remaining limitations in the QA report.
- README covers audience, prerequisites, exact installation/start/test commands, hosting, calculation assumptions, AI tool and verified model identity, and any human edits. Do not guess model identity or claim human edits.
- Transcript includes all accessible user-visible messages through delivery preparation, without summarisation. Redact credentials and explicitly identify any unavailable history; exclude hidden reasoning and system instructions.
- Produce a ZIP containing the root package. Exclude credentials, dependencies and temporary review assets.

Implementation and verification

1. Freeze the concise requirements and calculation assumptions before development.
2. Implement a responsive, keyboard-accessible page and a pure calculation module. Use a minimal local server, Node’s built-in unit test runner, and Playwright for critical browser automation.
3. Have an independent review agent examine the finance logic, validation, accessibility and packaging. Have a separate QA agent test against the story acceptance criteria.
4. Cover 10-, 15- and 20-year loans; maximum principal; property boundaries; invalid inputs; down-payment arithmetic; payment 120/121 transition; editable KIBOR; annual aggregation; final zero balance; and recalculation without stale results.
5. Verify numeric results against independently calculated reference cases. Test the critical flow on mobile and desktop widths.
6. Fix critical and high-severity defects, rerun affected checks and smoke tests, and record evidence. Mark stories verified only after their acceptance criteria pass.
7. Target calculation under 100 ms and initial application assets below 100 KB uncompressed, excluding documents. Record measurements and test environment.

Estimate potential national reach from official household, connectivity and scheme data, distinguishing eligible borrowers from visitors. Use explicit low/base/high adoption scenarios rather than claiming a precise user count. CDN-served assets and browser calculations avoid per-calculation server load; document hosting limits separately from measured performance.

Publishing and efficiency

- Create the public repository `TaimoorAliMangi/Taimoor-Ali-calculator`; verify creation permissions during implementation. Include source, documentation and the redacted transcript.

- Use the Sites workflow to deploy a public static application, verify deployment completion and anonymous access, and provide the live URL.
- A purchased domain is optional. Provide current registration and renewal comparisons for [Cloudflare Registrar](https://domains.cloudflare.com/) and [Porkbun](https://porkbun.com/products/domains); do not purchase anything.
- Work in four bounded passes: requirements, implementation, review/QA, and documents/publication. Share milestone updates and consolidate feedback before regenerating PDFs.
- Avoid repeated research, cosmetic iterations and duplicate tests. Usage consumption cannot be guaranteed within a percentage; keep scope focused on the agreed acceptance criteria.

10 User - 2026-10-13 00:27:21 PKT

things being discussed in the questions, add that as assumptions as well.

11 Assistant - 2026-10-13 00:27:55 PKT

Pakistan Housing Loan Calculator MVP

Summary

Build a lightweight English calculator for prospective subsidised housing borrowers. Publicly host it and deliver source code, a ZIP, a GitHub repository, product PDFs and the complete user-visible chat transcript.

Use plain HTML, CSS and JavaScript with browser calculations. Apply Pakistan government-inspired green and white styling, with clear independent-app identification.

Confirmed assumptions and decisions

Record these in the README and Design Report, including the choices discussed in our questions:

```
| Topic | Agreed assumption |
|---|---|
| Property limits | Inclusive: flats up to 1,500 sq ft and plots up to 10 marla. |
| Tenure | Any whole year from 10-20, selected through a dropdown. |
| Amount input | Entered amount means loan principal, capped at PKR 1 crore. |
| Down payment | Defaults to 10%; determines required equity and implied project cost. Changing it does not change repayments when loan principal stays fixed. |
| Future KIBOR | Editable assumption initialized from a dated official one-year KIBOR source; held constant after year 10 for estimation. |
| Language | English only for the MVP. |
| GitHub | Public repository named `Taimoor-Ali-calculator` in the connected personal account. |
| Transcript | Publish the full accessible user-visible session in GitHub and the deliverable package, with credentials redacted. |
| Review and QA | Separate, bounded review and QA agents after implementation. |
| Hosting | Public static hosting through Sites; a purchased domain is optional. |
```

Additional implementation assumptions: monthly reducing-balance repayment, no grace period, and full loan disbursement at the start. Exclude fees, insurance, taxes, prepayments and staged construction disbursements. Owned-plot equity is a simplified estimate requiring bank confirmation.

Calculator behavior

- Inputs: property type and size, loan principal, tenure, down-payment percentage and assumed KIBOR.
- Property types: flat, plot plus construction, and construction on an owned plot.
- Validate positive principal up to PKR 10,000,000, valid property boundaries, down payment from 10% to below 100%, and finite nonnegative KIBOR.
- Calculate implied project cost as `loan / (1 − down payment fraction)` and equity as project cost minus loan.
- Calculate monthly repayments at 5% annually over the full selected tenure. After payment 120, recalculate the remaining balance over the remaining months at assumed KIBOR plus three percentage points.
- Display prominent monthly payments for years 1-10 and the estimated later period, required equity, implied project cost and total repayment.
- Provide a yearly table containing year, monthly instalment, annual payments and year-end outstanding principal.
- Retain precision internally, round displayed PKR amounts and settle the final balance without an extra payment.
- Explain future-rate uncertainty and excluded costs beside the results.

Verify applicable scheme guidance before freezing requirements. The inclusive property limits, principal cap and 5% rate are supported by the [SBP March 2026 revision](https://www.sbp.org.pk/circulars/shsfd-circular-letter-no-01-of-2026).

Documents and package

Create `outputs/Taimoor-Ali-calculator/` containing:

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  jobs-to-be-done.pdf
  user-stories.pdf
  qa-report.pdf
transcripts/
  session-01.md
  session-01.pdf
```

- Research the prospective borrower persona and secondary family adviser and bank housing-finance adviser personas. Distinguish research-supported findings from assumptions.
- Use the requested role and jobs-to-be-done formats.
- Create one epic with features for eligibility inputs, calculation and results. Each story includes an ID, linked job, requested story format, Given/When/Then acceptance criteria and evidence-based status.
- Clone the retained Design Report template and preserve its visual system. Include executive summary, findings, implications, recommendations and appendix.
- Include the confirmed assumptions above, calculation conventions, national reach estimates and capacity assumptions in the report.
- Document review findings, requirements traceability, smoke/critical testing, automation results and

remaining limitations.
- README includes audience, prerequisites, exact install/start/test commands, hosting, assumptions, AI tool and verified model identity, and any human edits. Do not guess unavailable information.
- Transcript preserves all accessible user-visible messages through delivery preparation without summarisation. Identify unavailable history and credential redactions; exclude hidden reasoning and system instructions.
- Render and inspect every PDF page. Produce a ZIP of the root package, excluding credentials, installed dependencies and temporary assets.

Build, QA and publication

1. Freeze concise requirements and assumptions before development.
2. Implement a responsive, keyboard-accessible page and pure calculation module, with a minimal local server.
3. Add Node unit tests and Playwright automation for critical flows.
4. Use independent review and QA agents to assess code and story acceptance criteria.
5. Cover 10-, 15- and 20-year loans, maximum principal, property boundaries, invalid inputs, down-payment arithmetic, payment 120/121 transition, KIBOR changes, annual aggregation, final balance and stale-result prevention.
6. Check numerical results against independently calculated reference cases and test mobile/desktop layouts.
7. Fix critical and high-severity defects; rerun affected checks and smoke tests. Mark stories verified only after acceptance criteria pass.
8. Target calculation below 100 ms and application assets below 100 KB uncompressed, excluding documents; record measurements.
9. Estimate national reach using official household, connectivity and scheme data. Separate potential borrowers from website visitors and document low/base/high adoption assumptions. Do not claim untested traffic capacity.
10. Create the public GitHub repository, synchronize the package, deploy through Sites and verify anonymous public access. Provide repository, live-app and ZIP links.

A domain is optional. Compare current registration and renewal costs from [Cloudflare Registrar](https://domains.cloudflare.com/) and [Porkbun](https://porkbun.com/products/domains), without purchasing.

Keep work to four bounded passes: requirements, implementation, review/QA, and documents/publication. Consolidate feedback before regenerating PDFs. Minimize repeated research and cosmetic iterations; account-wide usage consumption cannot be guaranteed within a percentage.

12 User - 2026-10-13 00:27:55 PKT

Also I want you to do a security review of the app so that there are no major security loop holes

13 Assistant - 2026-10-13 00:28:31 PKT

Pakistan Housing Loan Calculator MVP

Summary

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Use plain HTML, CSS and JavaScript with browser calculations. Apply Pakistan government-inspired

green and white styling, with clear independent-app identification.

Confirmed assumptions

Record these in the README and Design Report:

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| Amount input | Loan principal, capped at PKR 1 crore. |
| Down payment | Defaults to 10%; determines equity and implied project cost. Repayments remain
unchanged when principal stays fixed. |
| Future KIBOR | Editable, initialized from a dated official one-year KIBOR source and held
constant after year 10 for estimation. |
| Language | English only. |
| GitHub | Public `Taimoor-Ali-calculator` repository in the connected personal account. |
| Transcript | Full accessible user-visible session published with credentials redacted. |
| Review and QA | Separate, bounded review and QA agents. |
| Security | Security review is a release requirement; no known unresolved critical or
high-severity findings at publication. |
| Hosting | Public static hosting through Sites; a purchased domain is optional. |
```

Use monthly reducing-balance repayments and full initial disbursement. Exclude grace periods, fees, insurance, taxes, prepayments and staged construction disbursements. Treat owned-plot equity as a simplified estimate requiring bank confirmation.

Calculator behavior

- Inputs: property type and size, loan principal, tenure, down-payment percentage and assumed KIBOR.
- Property types: flat, plot plus construction, and construction on an owned plot.
- Validate positive principal up to PKR 10,000,000, inclusive property limits, down payment from 10% to below 100%, and finite nonnegative KIBOR.
- Calculate project cost as `loan / (1 − down payment fraction)` and equity as project cost minus loan.
- Calculate monthly repayments at 5% annually over the full selected tenure. After payment 120, recalculate the remaining balance over the remaining months at assumed KIBOR plus three percentage points.
- Display prominent monthly payments for the subsidised and later periods, equity, implied project cost and total repayment.
- Provide yearly rows showing monthly instalment, annual payments and year-end outstanding principal.
- Retain precision internally, round displayed PKR amounts and settle the final balance without an extra payment.
- Explain future-rate uncertainty and excluded costs beside results.

Verify applicable scheme guidance before freezing requirements. The inclusive sizes, principal cap and 5% rate are supported by the [SBP March 2026 revision](https://www.sbp.org.pk/circulars/shsfd-circular-letter-no-01-of-2026).

Documents and package

Create `outputs/Taimoor-Ali-calculator/` containing:

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```

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  design-report.pdf
  app-roles.pdf
  jobs-to-be-done.pdf
  user-stories.pdf
  qa-report.pdf
  security-review.md
  security-review.pdf
transcripts/
  session-01.md
  session-01.pdf
```

- Research prospective borrower, family adviser and bank housing-finance adviser personas; distinguish evidence from assumptions.
- Use the requested role, jobs-to-be-done and user-story formats.
- Create one epic with features for eligibility inputs, calculation and results. Each story has an ID, linked job, Given/When/Then criteria and evidence-based status.
- Clone the retained Design Report template and preserve its visual system. Include executive summary, findings, implications, recommendations and appendix.
- Document confirmed assumptions, calculation conventions, national reach estimates, review findings and requirements-to-test traceability.
- README covers audience, prerequisites, exact install/start/test commands, hosting, assumptions, verified AI tool/model information and any human edits.
- Preserve all accessible user-visible session messages through delivery preparation without summarisation. Identify unavailable history and credential redactions; exclude hidden reasoning and system instructions.
- Render and inspect every PDF page. ZIP the root package without credentials, installed dependencies or temporary assets.

Implementation, QA and security

1. Freeze concise requirements and assumptions, including security acceptance criteria.
2. Implement a responsive, keyboard-accessible page and pure calculation module. Use a minimal local server, Node unit tests and Playwright critical-flow automation.
3. Independently review code and test story acceptance criteria using separate review and QA agents.
4. Verify 10-, 15- and 20-year loans, maximum principal, property boundaries, invalid inputs, equity arithmetic, payment 120/121 transition, KIBOR changes, annual aggregation, final balance and stale-result prevention. Check numerical results independently and test mobile/desktop layouts.
5. Review security before publication:
   - Examine browser code, dependencies, configuration, deployment and public package.
   - Use safe DOM rendering; prohibit `eval`, dynamic code execution and insertion of user input as HTML.
   - Test injection strings, malformed values and excessive input lengths. Ensure calculation loops remain bounded by valid tenure.
   - Keep calculations local; introduce no accounts, database, personal-data collection or third-party tracking.
   - Scan the repository, ZIP, documents and transcript for credentials and unintended private content.

  - Check dependency advisories and lock dependency versions. Keep production dependencies minimal.
  - Configure and verify HTTPS and supported security headers, including a restrictive Content Security Policy, MIME sniffing protection, referrer policy and framing restrictions.
  - Ensure deployment exposes only intended application assets, not transcripts, documentation, development files or credentials.
6. Record security findings with severity, evidence, remediation and retest status. Fix all critical and high findings before release; document remaining lower-severity findings and platform limitations. Report the review’s scope without claiming absolute security.
7. Rerun affected tests and smoke checks after fixes. Mark stories verified only when their acceptance criteria pass.
8. Target calculations below 100 ms and application assets below 100 KB uncompressed, excluding documents. Record actual measurements.

Publication and efficiency

- Estimate national reach from official household, connectivity and scheme data. Separate potential borrowers from visitors and document low/base/high adoption assumptions.
- Use CDN-served static assets and local calculations. Document hosting limits separately from measured performance.
- Create and synchronize the public GitHub repository, deploy through Sites, verify anonymous access and production security controls, and provide repository, live-app and ZIP links.
- A domain is optional. Compare current registration and renewal costs from [Cloudflare Registrar](https://domains.cloudflare.com/) and [Porkbun](https://porkbun.com/products/domains); do not purchase.
- Work in four bounded passes: requirements, implementation, review/QA/security, and documents/publication. Consolidate feedback before regenerating PDFs.
- Minimize repeated research and cosmetic iterations. Account-wide usage consumption cannot be guaranteed within a percentage.

14 User - 2026-10-13 00:30:36 PKT

I want you to execute userstories, show progress in chat. Run app in local. Once I sign off. Host it in cloud as well. Do basic level security infrastructure review in code as well so that there are no code loop holes that we can get hacked. mange entire work using agents and sub agents. Create UI mocks as well first and once I approve, then proceed. Vet entire code as dev lead as well

15 Assistant - 2026-10-13 00:31:05 PKT

We’re still in **Plan mode**, so implementation cannot start yet. Switch to execution mode to begin with UI mocks. Your approval gates will be: **approve mocks → build and review locally → approve local app → publish to cloud**.

Calculator delivery with approval gates

Assumptions and scope

Retain all previously agreed requirements: English UI, inclusive property limits, principal capped at PKR 1 crore, 10-20-year tenure, 10% default down payment, editable dated KIBOR assumption, monthly reducing-balance calculation, annual breakdown, public GitHub package and redacted user-visible transcript.

Use lightweight browser-based calculations, no accounts or database, and Pakistan government-inspired styling. Document all assumptions in the README and Design Report.

Cloud publication requires explicit local-app sign-off. UI implementation requires explicit mock approval.

Stage 1 - Requirements and UI mocks

- Create personas, app roles, jobs to be done, epic/features and user stories with linked jobs and Given/When/Then acceptance criteria.
- Prepare desktop and mobile UI mocks showing inputs, validation, prominent instalments and yearly results.
- Include the year-11 rate change, equity estimate and calculation limitations.
- Present mocks in chat for approval. Resolve feedback before implementing the app.

Stage 2 - Execute stories and run locally

- Coordinate bounded implementation, review and QA agents; keep overall integration and acceptance decisions with the lead agent.
- Implement approved stories and report progress in chat using story IDs and statuses: planned, in progress, under review, verified or blocked.
- Start the local app and provide its preview for user testing.
- Update the session transcript throughout the work.

Stage 3 - Dev-lead review, QA and security

- Independently review the entire application, calculation logic, dependencies, configuration and packaging as dev lead.
- Run unit tests against independent financial reference cases and browser automation against critical story acceptance criteria.
- Test property and amount boundaries, invalid inputs, equity arithmetic, rate transition, yearly totals, final balance, recalculation, keyboard access and mobile layout.
- Perform a basic code and infrastructure security review covering injection, safe DOM rendering, bounded inputs/calculations, dependency advisories, secret exposure, deployment configuration and unintended public files.
- Verify HTTPS and available security headers on the eventual hosted deployment.
- Fix and retest all critical and high-severity findings. Document remaining findings and review limitations; do not claim absolute protection against hacking.
- Present the tested local app, review summary and known limitations for sign-off. Keep cloud deployment pending.

Stage 4 - Package and publish after sign-off

- Create the requested Markdown documents and PDFs, using the retained Design Report template and inspecting every rendered page.
- Include QA and security reports, complete accessible user-visible transcript, and README with exact setup/start/test commands, AI tool/model details, human edits and assumptions.
- Package `Taimoor-Ali-calculator/` with `src/`, `tests/`, `docs/` and `transcripts/`; provide its ZIP.
- Create and synchronize the public GitHub repository.
- After explicit local-app approval, publish through Sites, verify anonymous access and production security configuration, and provide the live URL.
- Document evidence-based national reach scenarios and hosting capacity assumptions. Provide optional domain purchasing guidance without buying a domain.

Efficiency and acceptance

Use small, clearly assigned agent tasks, consolidated feedback and focused regression tests. Avoid

unnecessary frameworks, runtime AI calls and repeated document generation.

Completion requires approved mocks, verified user stories, passing critical automation, no unresolved critical/high review findings, inspected deliverables, synchronized GitHub source and-after sign-off-a verified public deployment.

16 User - 2026-10-13 00:31:12 PKT

PLEASE IMPLEMENT THIS PLAN:
Calculator delivery with approval gates

Assumptions and scope

Retain all previously agreed requirements: English UI, inclusive property limits, principal capped at PKR 1 crore, 10-20-year tenure, 10% default down payment, editable dated KIBOR assumption, monthly reducing-balance calculation, annual breakdown, public GitHub package and redacted user-visible transcript.

Use lightweight browser-based calculations, no accounts or database, and Pakistan government-inspired styling. Document all assumptions in the README and Design Report.

Cloud publication requires explicit local-app sign-off. UI implementation requires explicit mock approval.

Stage 1 - Requirements and UI mocks

- Create personas, app roles, jobs to be done, epic/features and user stories with linked jobs and Given/When/Then acceptance criteria.
- Prepare desktop and mobile UI mocks showing inputs, validation, prominent instalments and yearly results.
- Include the year-11 rate change, equity estimate and calculation limitations.
- Present mocks in chat for approval. Resolve feedback before implementing the app.

Stage 2 - Execute stories and run locally

- Coordinate bounded implementation, review and QA agents; keep overall integration and acceptance decisions with the lead agent.
- Implement approved stories and report progress in chat using story IDs and statuses: planned, in progress, under review, verified or blocked.
- Start the local app and provide its preview for user testing.
- Update the session transcript throughout the work.

Stage 3 - Dev-lead review, QA and security

- Independently review the entire application, calculation logic, dependencies, configuration and packaging as dev lead.
- Run unit tests against independent financial reference cases and browser automation against critical story acceptance criteria.
- Test property and amount boundaries, invalid inputs, equity arithmetic, rate transition, yearly totals, final balance, recalculation, keyboard access and mobile layout.
- Perform a basic code and infrastructure security review covering injection, safe DOM rendering, bounded inputs/calculations, dependency advisories, secret exposure, deployment configuration and unintended public files.
- Verify HTTPS and available security headers on the eventual hosted deployment.
- Fix and retest all critical and high-severity findings. Document remaining findings and review

limitations; do not claim absolute protection against hacking.
- Present the tested local app, review summary and known limitations for sign-off. Keep cloud deployment pending.

Stage 4 - Package and publish after sign-off

- Create the requested Markdown documents and PDFs, using the retained Design Report template and inspecting every rendered page.
- Include QA and security reports, complete accessible user-visible transcript, and README with exact setup/start/test commands, AI tool/model details, human edits and assumptions.
- Package `Taimoor-Ali-calculator/` with `src/`, `tests/`, `docs/` and `transcripts/`; provide its ZIP.
- Create and synchronize the public GitHub repository.
- After explicit local-app approval, publish through Sites, verify anonymous access and production security configuration, and provide the live URL.
- Document evidence-based national reach scenarios and hosting capacity assumptions. Provide optional domain purchasing guidance without buying a domain.

Efficiency and acceptance

Use small, clearly assigned agent tasks, consolidated feedback and focused regression tests. Avoid unnecessary frameworks, runtime AI calls and repeated document generation.

Completion requires approved mocks, verified user stories, passing critical automation, no unresolved critical/high review findings, inspected deliverables, synchronized GitHub source and-after sign-off-a verified public deployment.

17 Assistant - 2026-10-13 00:31:16 PKT

I’ll start with the requirements and desktop/mobile UI mocks, using agents for those bounded tasks. I’ll show you the mocks for approval before building the calculator. Cloud hosting will wait until you sign off on the tested local app.

18 Assistant - 2026-10-13 00:32:36 PKT

The requirements, mock design and security checklist are being prepared in parallel. I’ve checked the mock’s example: a PKR 50 lakh loan over 20 years gives about PKR 32,998/month initially, rising to PKR 50,193 from year 11 if KIBOR is assumed to be 12%. The mock will label that rate as illustrative.

19 Assistant - 2026-10-13 00:34:00 PKT

Stage 1 is ready: personas, assumptions, seven user stories and a security checklist are drafted. All stories remain **planned, awaiting mock approval**.

Preview **Desktop estimate**, **Mobile estimate** and **Validation** below. Values are illustrative; this is not the working calculator.

Visual preview reference: /Users/taimoorali/.codex/visualizations/2026/10/12/01a1318f-1922-7c31-8d54-428456c43d0b/housing-calculator-mocks.html

Do you approve this design, or want changes? Your requested approval gate requires this sign-off before implementation. Once approved, I’ll build and run locally, then complete dev-lead review, QA and security checks.

20 User - 2026-10-13 00:36:58 PKT

Looks good. Add eligibility criteria as text as well as link to PM housing scheme homepage for more details.
eligibility criteria = CNIC+ 50K income + 18 and above etc.

Also add knowledgeCIty logo at the top of the page who developed this software

21 Assistant - 2026-10-13 00:37:02 PKT

I'll treat the design as approved with those additions. I'll verify the eligibility wording and official scheme link, add the text, and start the local build. I'll also look for the KnowledgeCity logo so the developer credit uses the correct branding.

22 Assistant - 2026-10-13 00:38:13 PKT

I found the [official scheme homepage](https://apnaghar.gov.pk/) and KnowledgeCity's official logo. I'll add both.

PKR 50,000 is not a universal income requirement: [Meezan lists PKR 40,000 gross monthly income and a minimum age of 18](https://www.meezanbank.com/apna-ghar-program/). I'll show CNIC/eligible overseas identity, first-home ownership and bank-specific age/income requirements, so the app doesn't incorrectly rule people out.

23 User - 2026-10-13 00:38:44 PKT

token usage at 80%. optimze rest

24 Assistant - 2026-10-13 00:38:49 PKT

I'll keep the remaining work lean: one implementation pass, focused critical tests, and concise review reports. I'll combine code/security review and keep browser QA separate, avoiding extra design work and repeated PDF generation.

25 Assistant - 2026-10-13 00:40:13 PKT

The approved build is underway. US1-US4 cover inputs and repayments; US5 adds eligibility and source links; US6-US7 cover accessibility and safe local calculations. The next milestone is the local preview, followed by independent review and critical QA.

26 Assistant - 2026-10-13 00:42:21 PKT

The calculator is running locally with the KnowledgeCity logo, eligibility guidance and official scheme link. US1-US6 are now under review; unit tests pass. QA is checking the critical browser flows, while the review agent checks code and local-server security. Cloud hosting remains pending your local sign-off.

27 Assistant - 2026-10-13 00:42:54 PKT

Critical browser QA has passed the repayment cases, property limits, invalid-input handling, recalculation and mobile layout checks. Local security checks also passed for blocked private-file paths and loopback-only access. I'm finishing two small review changes: visible source links and a limit on unusually long numeric inputs.

28 User - 2026-10-13 00:43:43 PKT

Taimoor-Ali-Calculator is the repo name in github. Use that

29 Assistant - 2026-10-13 00:43:55 PKT

I’ll use the exact GitHub repository name **Taimoor-Ali-Calculator**. The local app is ready for your review; independent code and local security checks found no unresolved critical or high-severity issues.

30 Assistant - 2026-10-13 00:45:22 PKT

Eight unit tests and the critical browser suite pass. The app’s assets total about 27 KB, and the local calculation benchmark is comfortably below the 100 ms target. I’m completing the final injection/input-length checks and visual check before handing the local version back for sign-off.

31 Assistant - 2026-10-13 00:45:48 PKT

Confirmed: GitHub name will be **Taimoor-Ali-Calculator**.

[Open the local app](http://127.0.0.1:4173/) with eligibility text, official scheme links and KnowledgeCity branding.

- Eight unit tests and critical browser checks pass.
- Independent code/security review found no unresolved critical or high issues.
- Cloud security checks remain pending deployment.

Please sign off on the local app when ready. Your requested approval gate keeps cloud publishing pending until then.

32 User - 2026-10-13 00:48:17 PKT

UAT Feedback:
Each role needs to have a dropdown and once you select specific role, page should be optimized for that role.

Amount field should be slider Min 10 lac , Max 100 lacks

Create a language icon at top to convert page into Urdu for Urdu users

Check in the code on github and publish on a different branch and dont merge yet.

eligibility should be at bottom

Also generate 3 documents as originally described and show me folder path where all source code is available

33 Assistant - 2026-10-13 00:48:25 PKT

I’ll plan these UAT changes against the current app. We’re back in Plan mode, so I can inspect and specify the updates now; editing files and pushing the branch will require execution mode. I’ll retain the three documented roles and keep the work focused.

34 Assistant questions - 2026-10-13 00:48:33 PKT

How should selecting a role change the page?

- Focused guidance and results (Recommended): Borrower emphasizes monthly payments; family adviser emphasizes equity and annual commitments; finance adviser emphasizes rate assumptions and schedule, with the same calculator.

- Separate feature sets: Give each role different tools; specify the extra capabilities needed before implementation.

35 User question responses - 2026-10-13 00:48:37 PKT

How should selecting a role change the page?
Answer: Focused guidance and results (Recommended)

36 Assistant - 2026-10-13 00:48:56 PKT

UAT updates and unmerged GitHub delivery

Changes

- Add a role dropdown: **Prospective borrower**, **Family adviser**, **Housing-finance adviser**. Default to borrower.
- Keep the same calculator and entered values across roles. Borrower view emphasizes monthly payments; family view emphasizes equity and annual commitments; finance-adviser view emphasizes rates and the schedule. Change guidance and result ordering accordingly.
- Replace the amount input with a keyboard-accessible slider: **PKR 1,000,000-10,000,000**, in **PKR 100,000 steps**, defaulting to PKR 5,000,000. Show the selected amount in PKR and lakh.
- Add a top language icon with a labelled English/Urdu switch. Translate all interface text, validation, eligibility and role guidance; apply right-to-left Urdu layout. Preserve inputs and results when switching. Keep KnowledgeCity branding and readable PKR numbers.
- Move eligibility below the calculator and results, retaining the official scheme and bank-source links.

Documents and source

Generate the three requested PDFs using the retained Design Report template’s visual system:

- **App roles:** required role wording and updated role-specific behavior.
- **Jobs to be done:** requested situation/motivation/outcome format.
- **User stories:** linked jobs, BDD acceptance criteria and actual verification status.

Keep their Markdown sources alongside the PDFs. Update README and assumptions for the slider, role views and Urdu support.

All source is currently in:

`/Users/taimoorali/Documents/Codex/2026-10-13/act-as-a-technical-product-manager/outputs/Taimoor-Ali-calculator/`

Verification and GitHub

- Use bounded implementation, QA and code/security review agents.
- Test slider limits and keyboard operation, each role’s presentation, English/Urdu switching, right-to-left mobile layout, translated errors and unchanged financial results.
- Rerun affected repayment and security checks; render and inspect every PDF page.
- Synchronize with **Taimoor-Ali-Calculator**, inspecting its existing contents before pushing.
- Push changes on **`uat-feedback`**, preserving the default branch. Do not merge.

- Present the updated local app, three PDFs, source-folder link and GitHub branch link.

Assumptions and gates

Role selection changes presentation and guidance; it does not grant permissions or introduce accounts. Language selection remains local to the page.

Your UAT instructions authorize these revisions without another mock gate. Cloud deployment remains pending your sign-off on the updated local app. Execution resumes after leaving Plan mode.

37 User - 2026-10-13 00:49:06 PKT

PLEASE IMPLEMENT THIS PLAN:
UAT updates and unmerged GitHub delivery

Changes

- Add a role dropdown: **Prospective borrower**, **Family adviser**, **Housing-finance adviser**. Default to borrower.
- Keep the same calculator and entered values across roles. Borrower view emphasizes monthly payments; family view emphasizes equity and annual commitments; finance-adviser view emphasizes rates and the schedule. Change guidance and result ordering accordingly.
- Replace the amount input with a keyboard-accessible slider: **PKR 1,000,000-10,000,000**, in **PKR 100,000 steps**, defaulting to PKR 5,000,000. Show the selected amount in PKR and lakh.
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- Synchronize with **Taimoor-Ali-Calculator**, inspecting its existing contents before pushing.
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Assumptions and gates

Role selection changes presentation and guidance; it does not grant permissions or introduce accounts. Language selection remains local to the page.

Your UAT instructions authorize these revisions without another mock gate. Cloud deployment remains pending your sign-off on the updated local app. Execution resumes after leaving Plan mode.

38 Assistant - 2026-10-13 00:49:10 PKT

I'll implement the UAT changes, verify the role views and Urdu layout, generate the three PDFs, and push to `uat-feedback` without merging. The app will remain local until you approve cloud publication.

39 Assistant - 2026-10-13 00:51:08 PKT

I checked the repository: `main` contains only the initial Git attributes file. The `uat-feedback` branch is created; `main` is unchanged. Implementation, QA/security review, and the three template-based documents are being handled in parallel.

40 Assistant - 2026-10-13 00:53:34 PKT

The role views, loan slider and Urdu switch are implemented locally. Role and language changes preserve the entered values and results; eligibility now appears below the calculator. QA is checking those behaviors and the Urdu mobile layout while the PDFs are rendered and inspected.

41 Assistant - 2026-10-13 00:55:19 PKT

Verification passed: eight unit tests and eleven browser test groups, including role ordering, slider keyboard control, Urdu errors and mobile layout, and unchanged repayment results. Local code/security review found no unresolved critical or high issues. I'm finishing the PDF checks before pushing the complete package to `uat-feedback`.

42 Assistant - 2026-10-13 00:58:45 PKT

The three PDFs have passed visual checks: App Roles, Jobs to Be Done, and User Stories with BDD criteria and verified local statuses. I'm uploading the reviewed source and documents to `uat-feedback`; nothing will be merged into `main`.

43 User - 2026-10-13 00:59:47 PKT

30% remaining

44 Assistant - 2026-10-13 00:59:59 PKT

I'll use the remaining budget to finish the branch upload and handoff only. The app, QA/security review and three PDFs are complete; no further feature work or cloud deployment is planned in this pass.

45 Assistant - 2026-10-13 01:02:41 PKT

The documents are saved in the source folder and the branch upload is progressing. The remaining step is to commit the uploaded files, update `uat-feedback`, and verify that `main` still points to its original commit.

46 Assistant - 2026-10-13 01:04:43 PKT

Implemented and pushed to [uat-feedback](https://github.com/TaimoorAliMangi/Taimoor-Ali-Calculator/tree/uat-feedback). **Not merged; `main` is unchanged.**

[Open updated local app](http://127.0.0.1:4173/). Eight unit tests and eleven browser test groups pass; review found no unresolved critical/high issues.

PDFs: [App Roles](/Users/taimoorali/Documents/Codex/2026-10-13/act-as-a-technical-product-manager/outputs/Taimoor-Ali-calculator/docs/app-roles.pdf) · [Jobs to Be Done](/Users/taimoorali/Documents/Codex/2026-10-13/act-as-a-technical-product-manager/outputs/Taimoor-Ali-calculator/docs/jobs-to-be-done.pdf) · [User Stories](/Users/taimoorali/Documents/Codex/2026-10-13/act-as-a-technical-product-manager/outputs/Taimoor-Ali-calculator/docs/user-stories.pdf)

[Source folder](/Users/taimoorali/Documents/Codex/2026-10-13/act-as-a-technical-product-manager/outputs/Taimoor-Ali-calculator/):
`/Users/taimoorali/Documents/Codex/2026-10-13/act-as-a-technical-product-manager/outputs/Taimoor-Ali-calculator/`

Cloud publication remains pending your sign-off.

47 User - 2026-10-13 01:04:43 PKT

transcript folder is missing from main folder. Can you copy AI chat transcript and create pdf and save that

48 Assistant - 2026-10-13 01:04:48 PKT

I'll add the `transcripts` folder with a Markdown copy of the user-visible conversation and a matching PDF. I'll check the available chat history first and note any limits on completeness.

49 User - 2026-10-13 01:05:27 PKT

transcripts/
  session-01.md

50 Assistant - 2026-10-13 01:06:45 PKT

I found the saved session history, including the earlier plans and question responses. I'll save it as `transcripts/session-01.md` and `transcripts/session-01.pdf`, excluding internal reasoning, system context and raw tool logs.

51 Assistant - 2026-10-13 01:08:17 PKT

The Markdown transcript is saved with 50 message and question-response entries. I'm using the PDF workflow to paginate and check the matching PDF, then I'll sync both files to `uat-feedback`.

52 User - 2026-10-13 01:08:36 PKT

Sign off looks good. Merge with Main

53 Assistant - 2026-10-13 01:08:53 PKT

Sign-off received. I'll include this approval in the transcript, commit the transcript files, and merge the

completed UAT branch into `main`.

Export check: 53 message/question-response entries; 0 credential-pattern redactions.